1 Models

1.1 Arrivals

Abstract `Arrival` interface defines stochastic envelope for arrivals:

$$P(A(s,t) > \alpha(t-s,\sigma)) < \varepsilon(\sigma)$$

Each arrival class must implement the following functions:

- $\alpha(t,\sigma)$: `curve_func(self)` $\rightarrow$ `Callable[..., float]`;
- $\varepsilon(\sigma)$: `err_func(self)` $\rightarrow$ `Callable[..., float]`.

1.1.1 Token bucket arrivals

Class `TokenBucketArrival` has rate r and burst b as parameters, and implements the following functions:

- $\alpha(t,\sigma) = r \cdot t + b + \sigma$;
- $\varepsilon(\sigma) = 0$.

1.1.2 Discrete Markov-modulated On-Off arrivals

Describes a MMOO EBB traffic. Starts at its steady state distribution.

Class `DiscreteMMOOArrival` has $stay_on$, $stay_off$ and $peak_rate$ as parameters, and implements the following functions:

- $\alpha(t,\sigma,\theta) = r(\theta) \cdot t + b(\theta) + \sigma$, where:
  - $r(\theta) = \frac{1}{\theta} \log\left(\frac{p_{00}+p_{11}e^{\theta P}+\sqrt{(p_{00}+p_{11}e^{\theta P})^2-4(p_{00}+p_{11}-1)e^{\theta P}}}{2}\right)$ with $stay_on = p_{11}$, $stay_off = p_{00}$ and $peak_rate = P$;
  - $b(\theta) = \frac{1}{\theta} \log\left(e^{\theta P} \cdot \frac{\max\limits_{k\in\overline{1,M}}\{v_k\}}{\min\limits_{k\in\overline{1,M}}\{v_k\}} \cdot \frac{2}{p_{00}+p_{11}e^{\theta P}+\sqrt{(p_{00}+p_{11}e^{\theta P})^2-4(p_{00}+p_{11}-1)e^{\theta P}}}\right)$ with v_k - k-th component of the eigenvector $\boldsymbol{v}$ of matrix $\varphi(\theta)T$ (diagonal matrix containing MGFs of the traffic in each state multiplied by transition probability matrix);
- $\varepsilon(\sigma,\theta) = e^{-\theta\sigma}$.

1.1.3 Aggregated arrivals

Class `AggregatedArrival` takes a sequence of n arrivals A_i with curves α_i and errors ε_i, and implements the following methods:

- $\alpha(t,\sigma) = \sum\limits_{i=1}^{n} \alpha_i(t,\sigma)$;
- $\varepsilon(\sigma) = \sum\limits_{i=1}^{n} \varepsilon_i(\sigma)$.

We follow the sub-additivity of probability here:

$$P\Big(\sum_{i=1}^{n} A_i(s,t) > \sum_{i=1}^{n} \alpha_i(t-s,\sigma)\Big) \leqslant \sum_{i=1}^{n} P(A_i(s,t) > \alpha_i(t-s,\sigma)) < \sum_{i=1}^{n} \varepsilon_i(\sigma)$$

1.2 Services

Abstract `Service` interface defines a stochastic envelope for a service:

$$P(D(t) < A * [\beta - \sigma]^+(t)) < \varepsilon(\sigma)$$

Each service class must implement the following functions:

- `curve_func(self)` $\rightarrow$ `Callable[..., float]`;
- `err_func(self)` $\rightarrow$ `Callable[..., float]`.

1.2.1 Constant-rate service

Class `ConstantRateService` has rate R and burst b (default: 0) as parameters, and implements the following functions:

- $\beta(t) = Rt + b$;
- $\varepsilon(\theta) = 0$.

1.2.2 Leftover service

Class `LeftoverService` takes `service` S with service curve β, `cross_arrival` A_2 with arrival curve α_2 and error function ε_{a_2}, and `slack_rate` δ, as well as some additional parameters for infinite series calculation, and provides a stochastic service envelope for leftover service $S_{l.o.}$ for foi A_1. It implements the following functions:

- $\beta_{l.o.}(t) = \beta(t) - \alpha_{2,-\delta}(t)$;
- $\varepsilon_{l.o.}(\sigma) = \sum_{k=0}^{+\infty} \varepsilon_{a_2}(\sigma + k\delta)$.

Notes on the implementation:

1. If the arrivals are deterministic, the calculated error is 0. If the arrivals are EBB, the calculated error is $\frac{e^{-\theta\sigma}}{1-e^{-\theta\delta}}$. Otherwise, the infinite sum is trimmed;
2. Current implementation does **not** check the stability condition:

$$\lim_{t\to+\infty} \sup\{\alpha_2(t) - \beta(t)\} < 0$$

3. Current implementation does **not** check the integrability condition for ε_{a_2}:

$$\int_0^{+\infty} \varepsilon_{a_2}(\sigma)d\sigma < +\infty$$